PRAKTIKUM BASIS DATA

MODUL 9

Scripting Procedure dan Function

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

MODUL 9

JURNAL

Running [code sql] berikut, screenshot hasilnya dan analisis hasil code tersebut

1. create table employees as select * from hremployees.
2. select * from employees.
3.

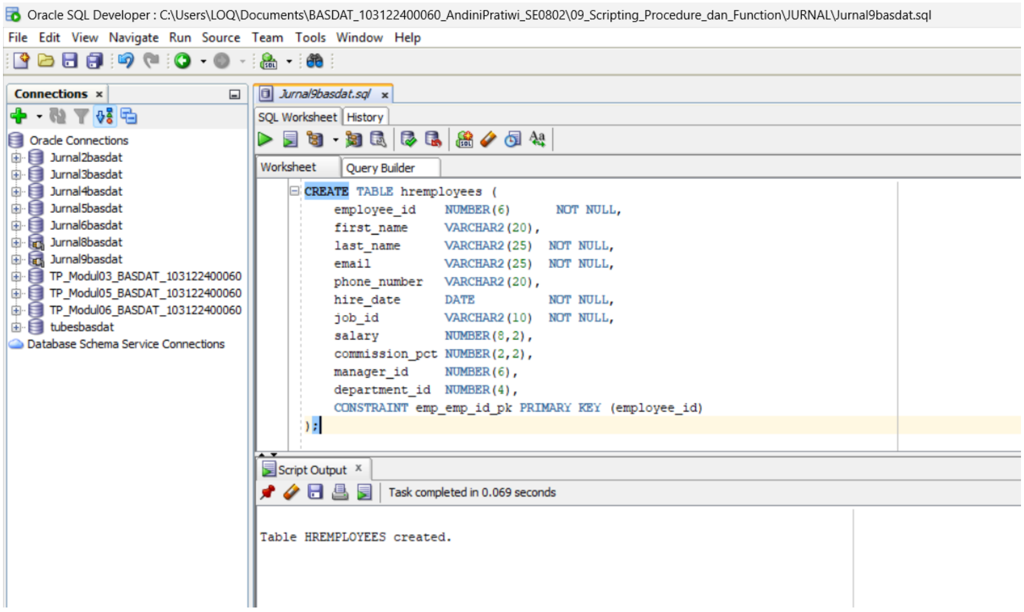
```
CREATE OR REPLACE PROCEDURE increase_salary (
department_id_in   IN employees.department_id%TYPE,
    increase_pct_in    IN NUMBER)
     IS
BEGIN
    FOR employee_rec
            IN (SELECT employee_id
            FROM employees
           WHERE department_id = increase_salary.department_id_in)
    LOOP
            UPDATE employees emp
                SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
            WHERE emp.employee_id = employee_rec.employee_id;
            DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;

```

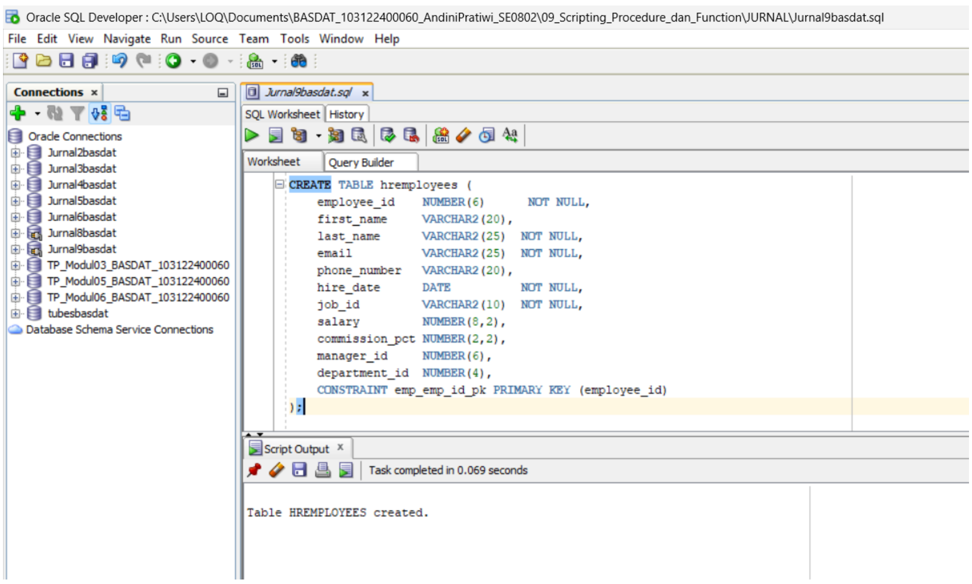
4. BEGIN
   increase_salary (90, .50);
   ROLLBACK;
   END;
   /
   select * from employees where department_id = 90
5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.

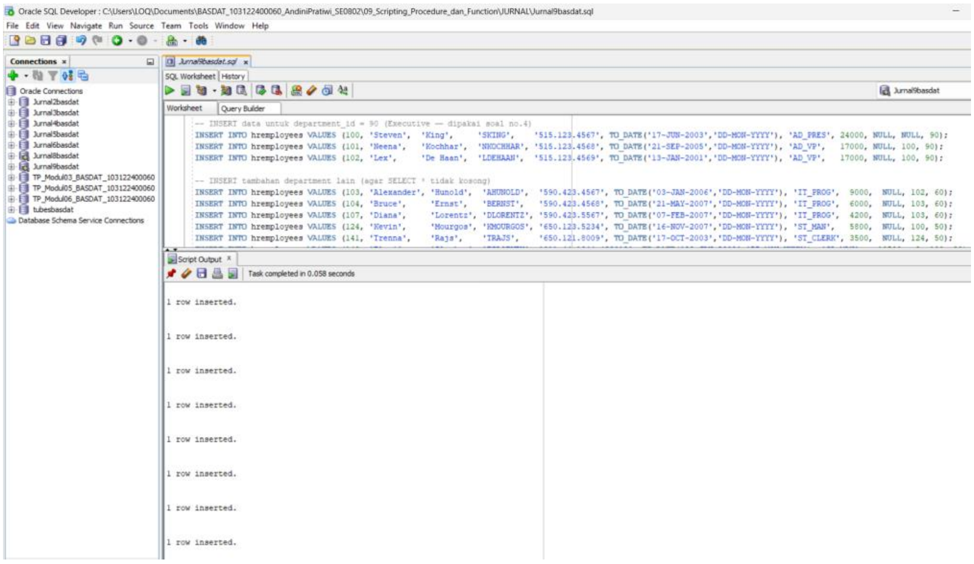
Jawaban:

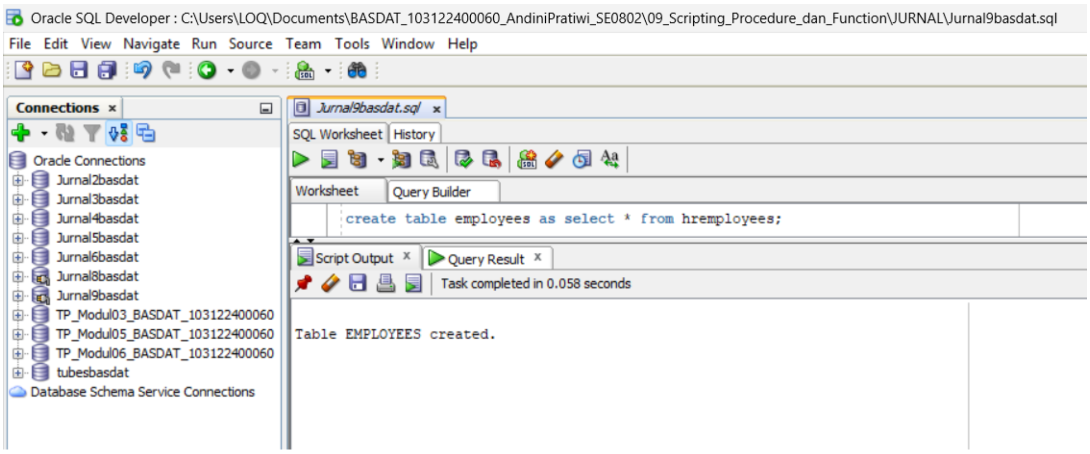
Running code sql berikut, screenshot hasilnya dan analisis hasil code tersebut



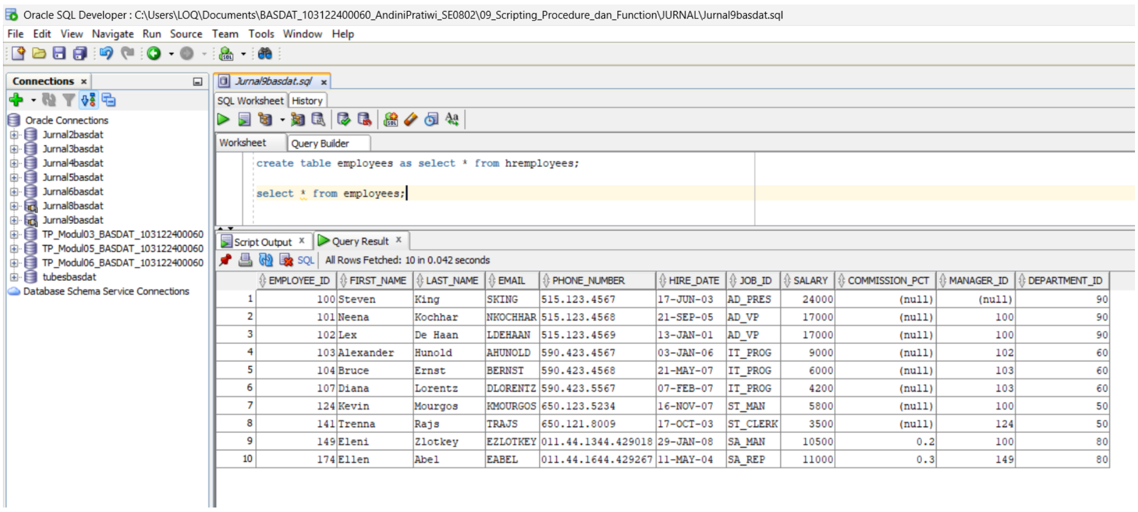
1. create table employees as select * from hremployees





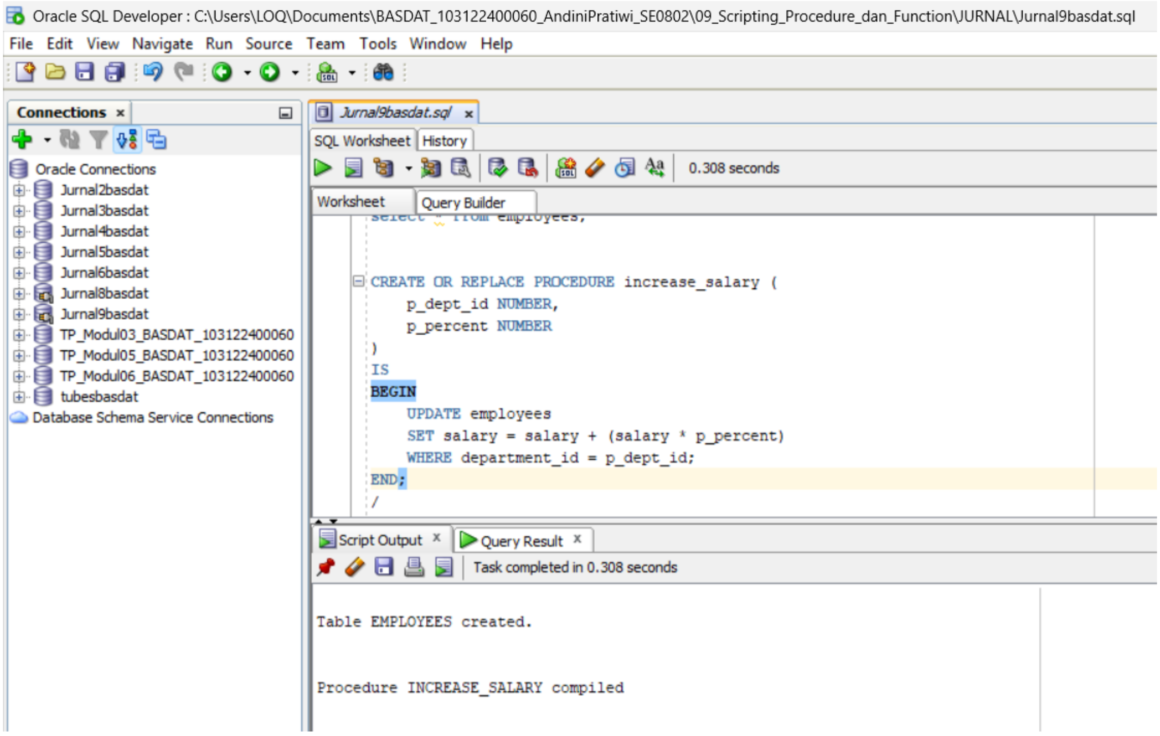


2. select * from employees



	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-OCT-03	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80

3.


Oracle SQL Developer : C:\Users\LOQ\Documents\BASDAT_103122400060_AndiniPratiwi_SE0802\09_Scripting_Procedure_dan_Function\JURNAL\Jurnal9basdat.sql
File Edit View Navigate Run Source Team Tools Window Help
Connections
Oracle Connections
Jurnal2basdat
Jurnal3basdat
Jurnal4basdat
Jurnal5basdat
Jurnal6basdat
Jurnal8basdat
Jurnal9basdat
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TP_Modul05_BASDAT_103122400060
TP_Modul06_BASDAT_103122400060
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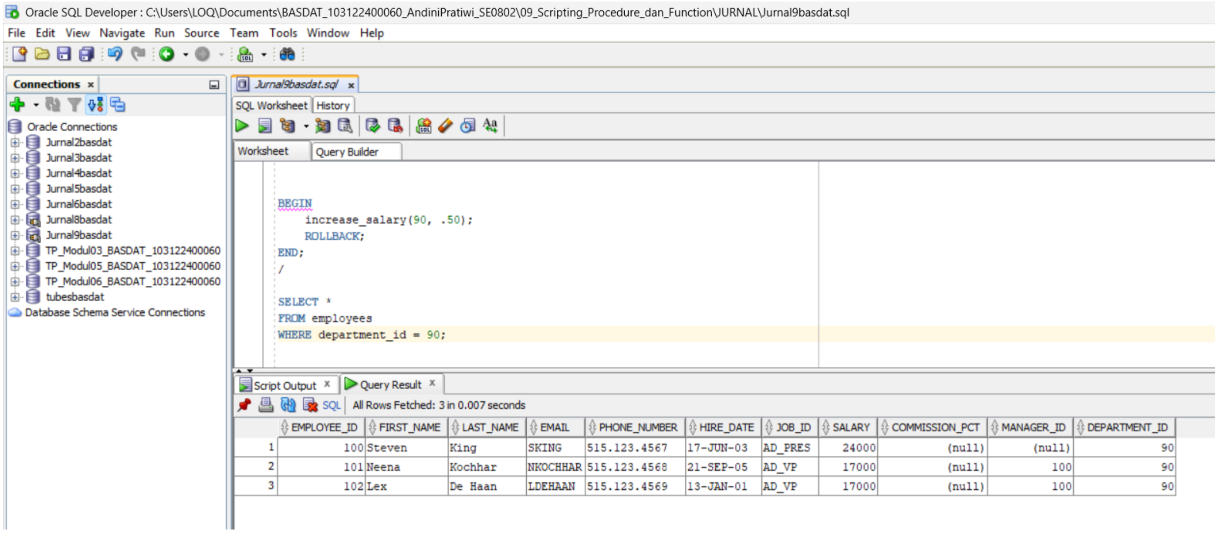
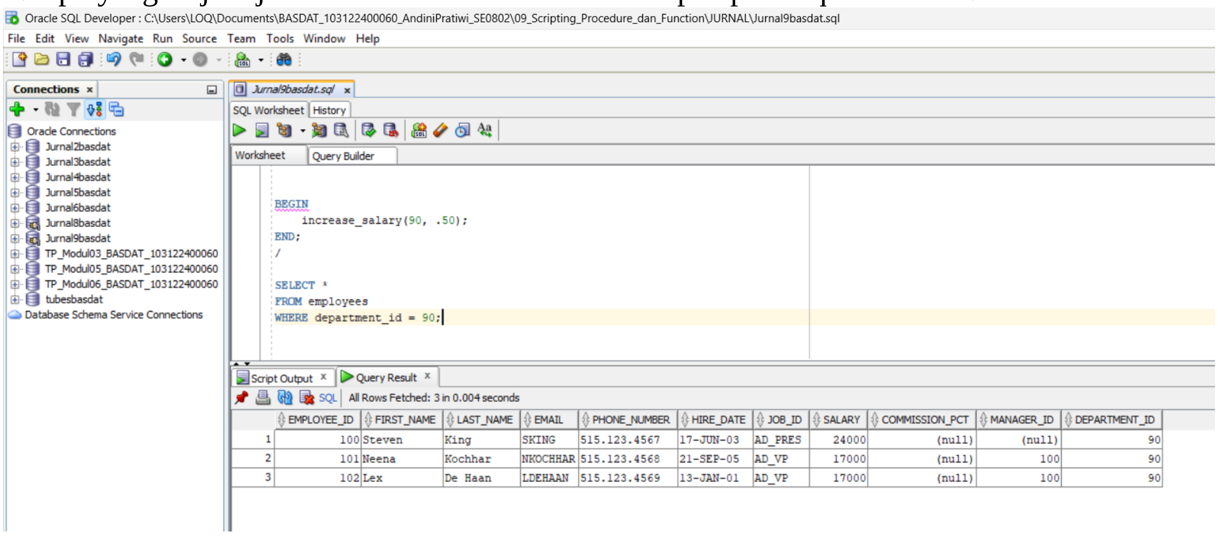
Database Schema Service Connections
Jurnal9basdat.sql
SQL Worksheet
History
0.308 seconds
Worksheet
Query Builder
CREATE OR REPLACE PROCEDURE increase_salary (
p_dept_id NUMBER,
p_percent NUMBER
)
IS
BEGIN
UPDATE employees
SET salary = salary + (salary * p_percent)
WHERE department_id = p_dept_id;
END;
/
Script Output
Query Result
Task completed in 0.308 seconds
Table EMPLOYEES created.
Procedure INCREASE_SALARY compiled

4. BEGIN increase_salary (90, .50); ROLLBACK; END; / select * from employees where department_id = 90



5. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.



Jika statement ROLLBACK; pada blok kode nomor 4 dihapus, maka perubahan kenaikan gaji sebesar 50% akan tersimpan pada tabel employees. Procedure increase_salary akan menaikkan salary seluruh pegawai yang berada pada department_id = 90 sebesar 50% dari gaji semula.
Saat ROLLBACK digunakan, perubahan yang dilakukan oleh procedure dibatalkan sehingga data kembali ke kondisi awal. Oleh karena itu hasil SELECT setelah procedure dijalankan tetap menampilkan nilai salary seperti sebelum dilakukan kenaikan.
Sebaliknya, jika ROLLBACK dihapus dan transaksi diakhiri dengan COMMIT atau tersimpan permanen, maka salary pegawai pada departemen 90 akan bertambah 50% dan perubahan tersebut akan terlihat pada hasil SELECT berikutnya.